

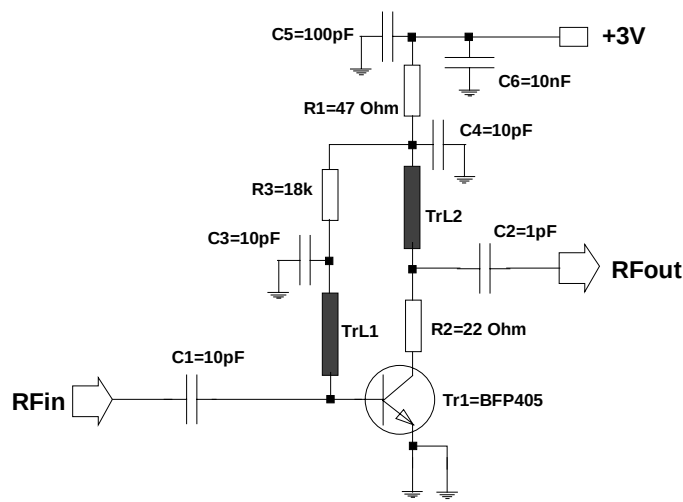
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A Low-Noise-Amplifier with good IP3out - performance at 1.9 GHz using BFP405

This application note describes a low noise amplifier at 1.9GHz using SIEMENS SIEGET[®]25 BFP405. The design emphasis has been on achieving high output intercept point. A circuit description, schematic, pcb layout and components list are shown below together with measured performance data.

Data at 1.9 GHz (3V and 8.5mA):

Gain=19.5 dB / IP3out=15dBm / NF=2.3dB / RLin-out >15dB



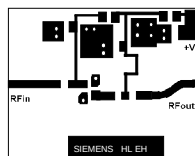
This amplifier at 1.9 GHz has been realized by using microstrip lines as matching elements. The design offers a good compromise between high IP3-values, high gain and return loss values.

In order to optimise the design for a particular application please observe the following points:

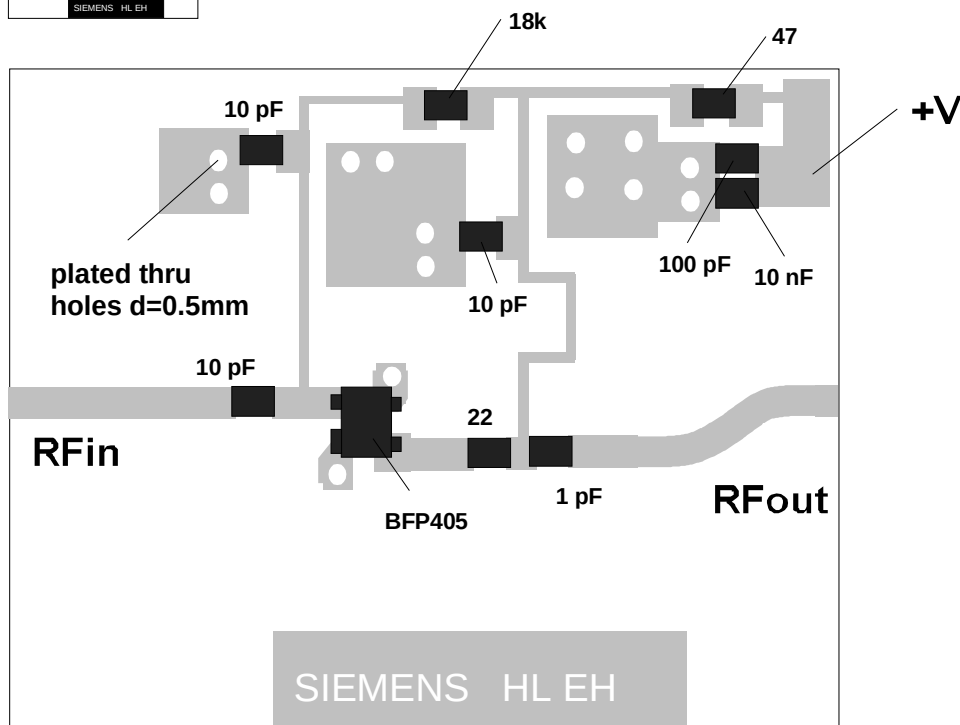
- The layout size can be reduced by using chip-coils instead of the microstrip lines TrL1 and TrL2.
- Improved stabilization behaviour versus temperature and reduced variation in amplifier performance due to the device's Beta (current gain) distribution can be achieved by using an active bias circuit. Such a circuit is available as a single device from Siemens - BCR400. For further information please refer to Application Note 014. However the resistors R1 and R3 are sufficient in most applications for stabilization purposes.
- This circuit is not optimized for low noise figure values. The measured figures include losses of SMA-connectors and the relatively high loss of the microstrip lines on the epoxy-board.
- The use of teflon material would provide an improvement of ≈ 0.1 dB.
- Resistor R2 is used to improve rf-circuit-stability and return loss values at the output. It also affects the output intermodulation performance.

Component	Value	Unit	Size	Comment
R1	47	Ω	0603	bias / collector-resistance / $V_{R1} \approx 0.5V$
R2	22	Ω	0603	to improve stability and output return loss
R3	18	$k\Omega$	0603	bias / base-resistor
C1	10	pF	0603	inputmatch
C2	1	pF	0603	outputmatch
C3	10	pF	0603	rf-short
C4	10	pF	0603	outputmatch
C5	100	pF	0603	rf-short
C6	10	nF	0603	rf-short
Tr1			SOT343	SIEGET BFP405
TrL1				inputmatch, w=0.3mm
TrL2				output-match, w=0.3mm
Substrate	FR4			h=0.5mm; $\epsilon_r=4.5$; 50 Ω micorstrip line w= 0.95mm

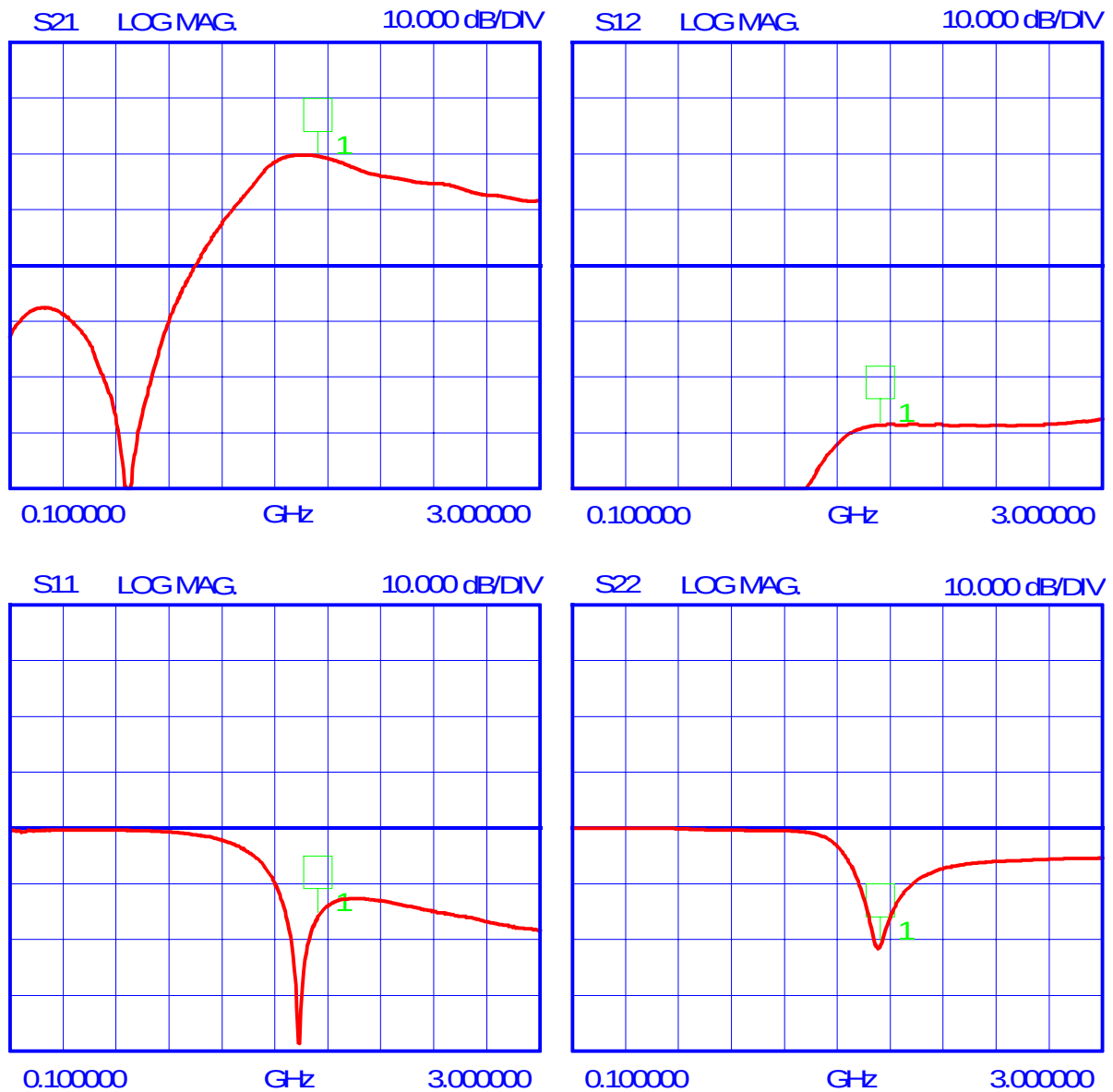
PCB Layout and Component Placement



scale 1:1
dim.: 25mm x 20mm



Measured data:



1 1.9 GHz

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